Structured-output tokenizer study

21 candidates • Modal CPU • 2026-10-16

18 existing tokenizers plus three fresh 64 MiB code-focused fits. Four workloads: Python excerpts, JSON, Git diffs and tool calls. 512 held-out source parents per workload; all candidates pass exact round-trip and zero-UNK checks.

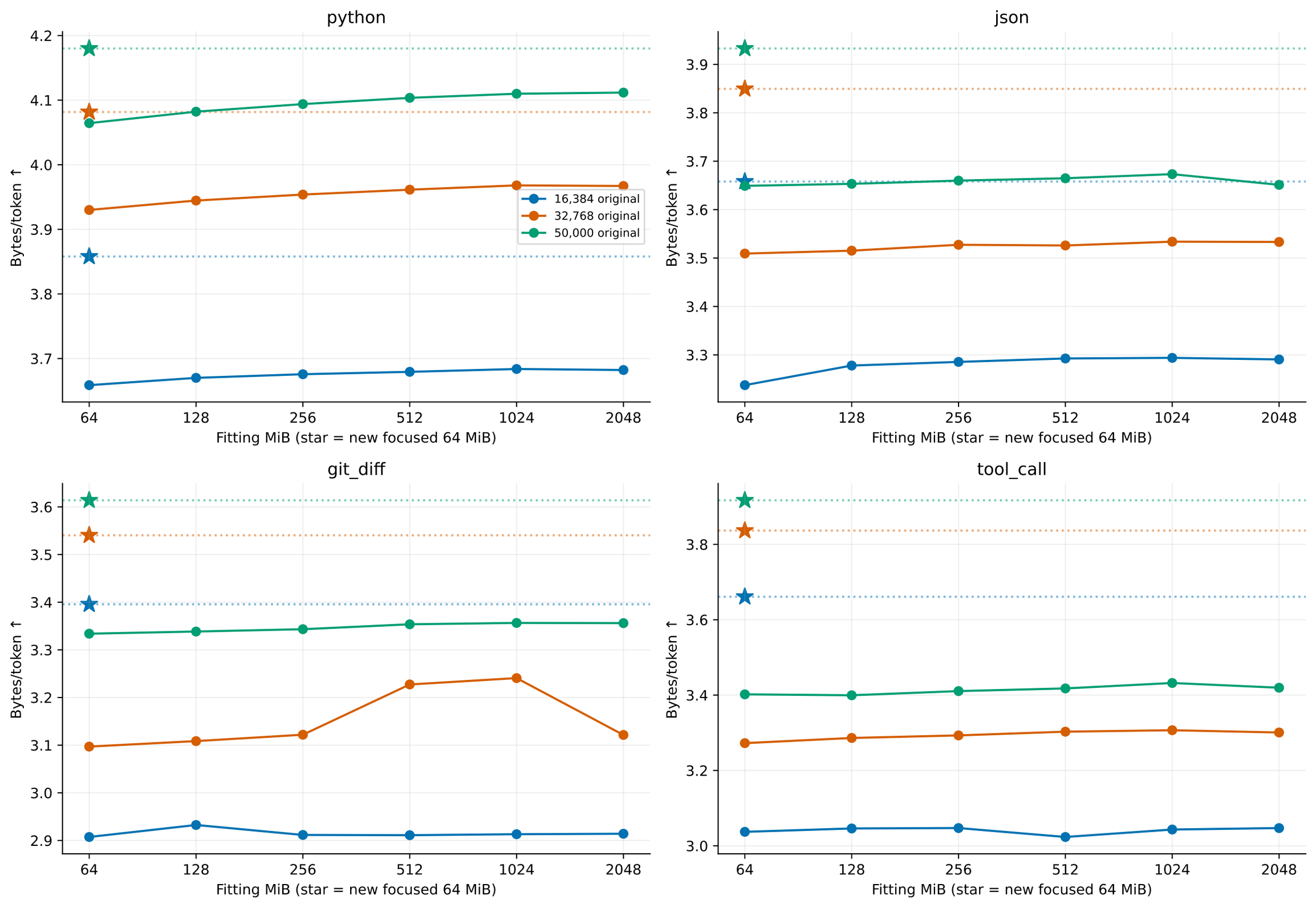
Focused fitting mixture: 40% Python, 20% JSON, 20% diffs and 20% tool-call formats by bytes. Same BPE rules and vocabulary sizes (16K, 32K, 50K).

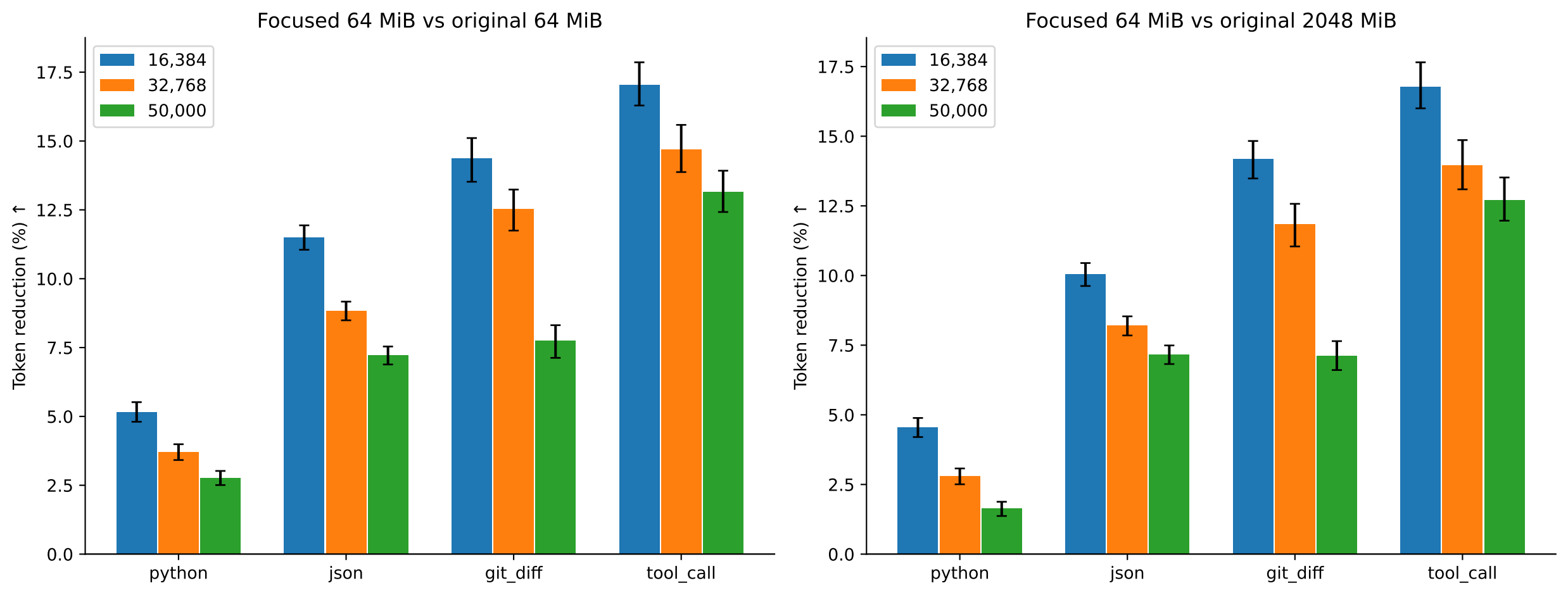
32K focused vs original 64 MiB: python: 3.72% fewer tokens; json: 8.84% fewer tokens; git_diff: 12.52% fewer tokens; tool_call: 14.70% fewer tokens.

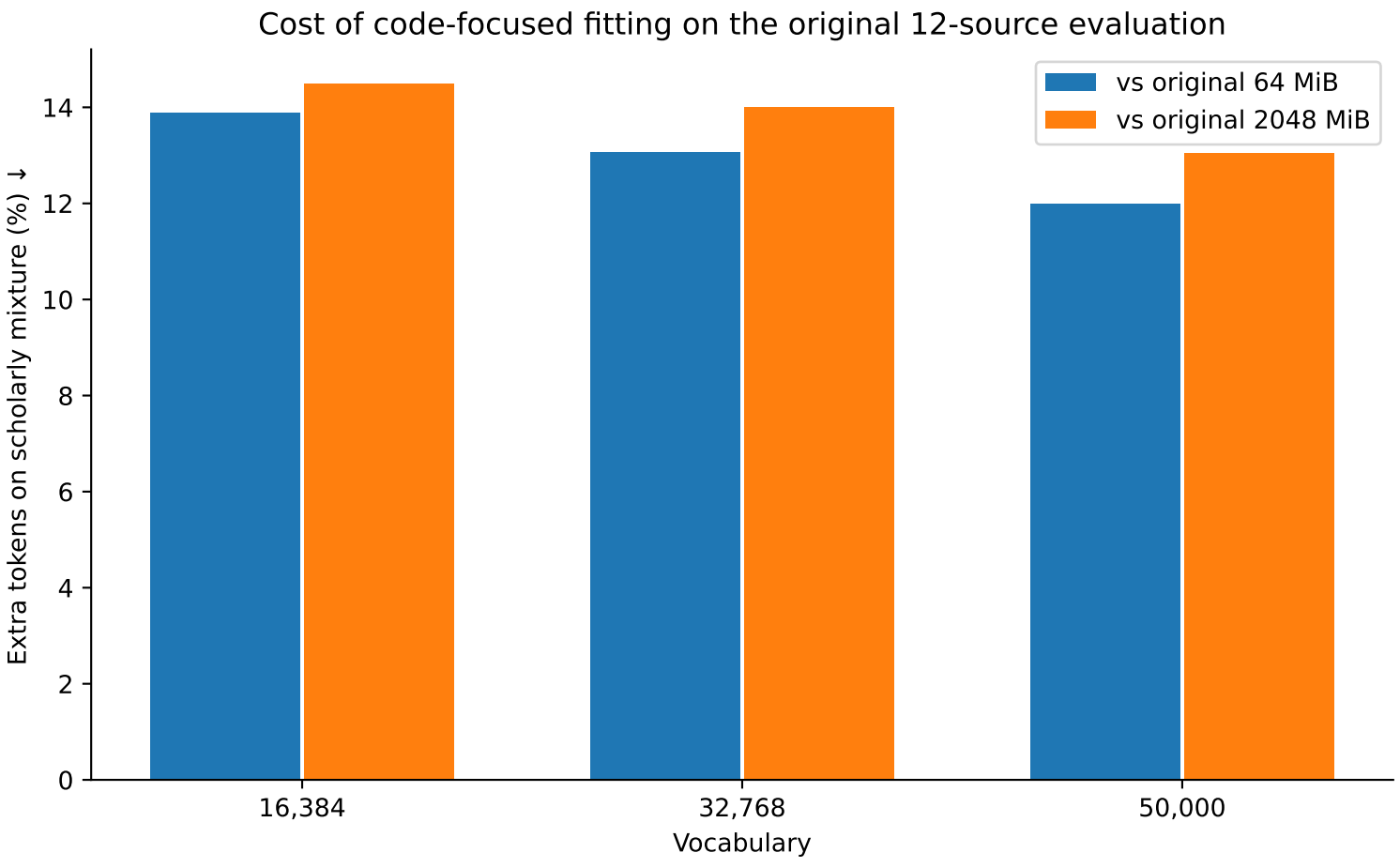
On the original scholarly mixture, the focused 32K tokenizer uses 13.07% more tokens than the original 64 MiB / 32K fit.

Caution: JSON, patches and tool calls are generated fixtures, with shared templates across disjoint source parents. This tests encoding efficiency, not production tool-call quality or model correctness. Existing published default is unchanged.

Full tables, methods and limitations are in REPORT.md; machine-readable results, paired intervals and artifact hashes accompany this PDF.


python
Bytes/token ↑
4.2
4.1
4.0
3.9
3.8
3.7
16,384 original
32,768 original
50,000 original
64
128
256
512
1024
2048
Fitting MiB (star = new focused 64 MiB)
json
Bytes/token ↑
3.9
3.8
3.7
3.6
3.5
3.4
3.3
64
128
256
512
1024
2048
Fitting MiB (star = new focused 64 MiB)
git_diff
Bytes/token ↑
3.6
3.5
3.4
3.3
3.2
3.1
3.0
2.9
64
128
256
512
1024
2048
Fitting MiB (star = new focused 64 MiB)
tool_call
Bytes/token ↑
3.8
3.6
3.4
3.2
3.0
64
128
256
512
1024
2048
Fitting MiB (star = new focused 64 MiB)


Focused 64 MiB vs original 64 MiB
Token reduction (%) ↑
16,384
32,768
50,000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
python
json
git_diff
tool_call
Focused 64 MiB vs original 2048 MiB
Token reduction (%) ↑
16,384
32,768
50,000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
python
json
git_diff
tool_call


Cost of code-focused fitting on the original 12-source evaluation
vs original 64 MiB
vs original 2048 MiB
Extra tokens on scholarly mixture (%) ↓
0
2
4
6
8
10
12
14
16,384
32,768
50,000
Vocabulary